

Software Engineering — 10-Page Master Quick Revision

High-Yield Formula Sheet, Metric Tables & Top 10 BIT Mesra PYQ Solutions

⚡ 10-Page Master Quick Revision — Software Engineering (CS24353)

Page 1: 67-Topic Master Syllabus Progress Checklist (Modules I – V)

Page 3: Process Models Comparison: Waterfall, Prototyping, Spiral & Agile

Page 5: Software Design Principles: Cohesion (7 Levels) vs. Coupling (5 Levels)

Page 7: Testing & V&V: Verification vs. Validation, White vs. Black-Box Testing

Page 9: Cost Estimation: Boehm's COCOMO Model & Albrecht's Function Points Math

Page 2: The Tier-S High-Yield Core Topic Ranking & Exam Strategy

Page 4: Requirements Engineering: Functional vs. Non-Functional & IEEE 830 SRS

Page 6: UML Architectural Diagrams: Use Case, Class, Activity & Sequence

Page 8: Basis Path Testing: Control Flow Graphs & McCabe's Cyclomatic Complexity

Page 10: SQA & Maintenance: CMMI Levels 1–5, SCM Change Control & CAP-P

The Complete 67-Topic Syllabus Master Checklist

67 / 67 Syllabus Topics Complete

Module	Topic #	Topic Name	Status
M1: Process & Planning (14)	1	What is Software Engineering? (Systematic Approach)	✔ Complete
	2	Definitions & The Software Crisis	✔ Complete
	3	The Evolving Role of Software (Product & Service)	✔ Complete
	4	Software Process Framework Activities	✔ Complete
	5	Waterfall Model (Sequential Phases)	✔ Complete
	6	Prototyping Model (Requirement Clarification)	✔ Complete
	7	Spiral Model (Risk Analysis Quadrants)	✔ Complete
	8	Rapid Application Development (RAD)	✔ Complete
	9	Incremental Model (Progressive Releases)	✔ Complete
	10	Agile Models (Scrum, Sprints & Manifesto)	✔ Complete
	11	Management Activities (The 4 P's Framework)	✔ Complete
	12	Project Planning (Scope, Cost & Resources)	✔ Complete
	13	Project Scheduling (WBS, Gantt, PERT/CPM)	✔ Complete
	14	Risk Management (I-A-P-M-M Cycle)	✔ Complete
M2: Requirements (12)	15	Functional Requirements (Behavioral Services)	✔ Complete
	16	Non-Functional Requirements (Quality Constraints)	✔ Complete
	17	User Requirements (Stakeholder Statements)	✔ Complete
	18	System Requirements (Detailed Technical Contracts)	✔ Complete
	19	Software Requirements Specification (SRS) Document	✔ Complete
	20	IEEE 830 Standard Format for SRS	✔ Complete
	21	9 Characteristics of Good SRS (C-U-C-C-V-F-T-P-M)	✔ Complete
	22	Feasibility Study (TELOS Framework)	✔ Complete
	23	Requirements Elicitation Techniques (Interviews, JAD)	✔ Complete
	24	Requirements Analysis & Conflict Resolution	✔ Complete
	25	Requirements Validation & Reviews	✔ Complete
	26	Requirements Management & Traceability Matrix	✔ Complete
M3: Design & UML (13)	27	Design Engineering (Architecture vs. Detail)	✔ Complete
	28	The Software Design Process	✔ Complete
	29	Software Design Quality Attributes	✔ Complete
	30	Fundamental Design Concepts (Abstraction & Hiding)	✔ Complete
	31	Design Models (Data, Architectural, Interface)	✔ Complete
	32	Object-Oriented Design (SOLID Principles)	✔ Complete

Module	Topic #	Topic Name	Status
	33	Cohesion (7 Levels: Coincidental to Functional)	✔ Complete
	34	Coupling (5 Levels: Content to Data Coupling)	✔ Complete
	35	UML Use Case Diagrams (Actors & Include/Extend)	✔ Complete
	36	UML Class Diagrams (Attributes, Methods & Relations)	✔ Complete
	37	UML Activity Diagrams (Workflows & Swimlanes)	✔ Complete
	38	UML Sequence Diagrams (Lifelines & Messages)	✔ Complete
	39	UML Collaboration / Communication Diagrams	✔ Complete
M4: Testing & V&V (16)	40	Verification ("Building the Product Right?")	✔ Complete
	41	Validation ("Building the Right Product?")	✔ Complete
	42	V&V Planning & Test Strategy	✔ Complete
	43	Software Inspection (Fagan Defect Reviews)	✔ Complete
	44	Static Analysis (Control & Data Flow Linters)	✔ Complete
	45	Software Testing Foundations & Defect Taxonomies	✔ Complete
	46	Testing Functions & Test Harness Design	✔ Complete
	47	Test Case Design Artifact Structure	✔ Complete
	48	White-Box Testing (Statement & Branch Coverage)	✔ Complete
	49	Black-Box Testing (Equivalence Partitioning & BVA)	✔ Complete
	50	Basis Path Testing & McCabe's Cyclomatic Complexity	✔ Complete
	51	Control Structure Testing (Condition & Loop)	✔ Complete
	52	Unit Testing (Stubs & Drivers)	✔ Complete
	53	Integration Testing (Top-Down, Bottom-Up, Sandwich)	✔ Complete
	54	System Testing (Performance, Stress, Recovery)	✔ Complete
	55	Software Reliability & Availability (MTBF, MTTR)	✔ Complete
M5: Estimation & Quality (13)	56	Process & Product Metrics (Defect Density)	✔ Complete
	57	Software Measurement (Lines of Code LOC)	✔ Complete
	58	Software Project Estimation Paradigms	✔ Complete
	59	Decomposition Techniques (Problem vs. Process)	✔ Complete
	60	Empirical Estimation Models ($\text{Effort} = a \text{Size}^b$)	✔ Complete
	61	The COCOMO Model (Organic, Semi, Embedded)	✔ Complete
	62	Function Point Analysis (5 Function Types: EI, EO, EQ, ILF, EIF)	✔ Complete
	63	Quality Assurance & Standards (ISO 9001 & CMMI 1–5)	✔ Complete
	64	Quality Planning & SQA Checklists	✔ Complete

Module	Topic #	Topic Name	Status
	65	Quality Control (QA vs. QC: Prevention vs. Detection)	✅ Complete
	66	Software Configuration Management (SCM & CCB)	✅ Complete
	67	Software Maintenance (The 4 CAP-P Maintenance Types)	✅ Complete
	68	Software Re-Engineering (Reverse & Forward Engineering)	✅ Complete

★ Tier-S Core Exam Topics (Must Master)

#	Topic	Core Concept / Formula
1	Process Models	Waterfall (Sequential) vs. Spiral (Risk) vs. Agile (Sprint)
2	IEEE 830 SRS	C-U-C-C-V-F-T-P-M Characteristics & 3-Section Format
3	Functional vs. Non-Functional	WHAT the system does vs. HOW WELL (Constraints & Performance)
4	Cohesion vs. Coupling	High Functional Cohesion (Internal focus) + Low Data Coupling (Inter-dependency)
5	UML Dynamic Modeling	Sequence (Time-oriented lifelines) & Activity (Swimlanes/Forks)
6	Verification vs. Validation	"Building product right?" (Static specs) vs "Building right product?" (Dynamic user needs)
7	White vs. Black-Box	Structural logic/branch coverage vs. Functional I/O (EP & BVA)
8	Cyclomatic Complexity	$V(G) = E - N + 2 = P + 1$ (Linearly independent basis paths)
9	Basic COCOMO	$\text{Effort} = a(\text{KLOC})^b$, $T_{\text{dev}} = c(E)^d$ (Organic, Semi-Detached, Embedded)
10	Function Point Analysis	$\text{FP} = \text{UFP} \times [0.65 + 0.01 \times \sum C_i]$ (EI, EO, EQ, ILF, EIF)
11	CMMI Levels 1-5	Initial → Managed → Defined → Quantitatively Managed → Optimizing
12	Software Maintenance	CAP-P: Corrective (20%), Adaptive (25%), Perfective (50%), Preventive (5%)